Lecture 7

Smooth manifolds

Objectives. *A smooth manifold has local coordinates whose changes are smooth. This allows derivatives to be compared between coordinate systems. We construct coordinates on spheres and projective spaces, then prove that regular constraints define manifolds using the implicit function theorem. The space of pure qubit states gives a concrete example: a sphere described by two complex coordinate charts.*

By the end of this lecture you will be able to:

- Check chart transitions and construct a smooth atlas.
- Test smooth maps in coordinates and apply the chain rule.
- Prove that a nonempty regular level set is an embedded submanifold.
- Relate the two projective qubit coordinates by $w = 1/z$.

7.1 Charts and smooth structures

A *topological n-manifold* is a Hausdorff, second-countable space M locally homeomorphic to open subsets of $\mathbb{R}^n$. Here second-countable means that its topology has a countable basis. A *chart* is a pair (U, φ), with $U \subseteq M$ open and $\varphi\colon U \to \varphi(U) \subseteq \mathbb{R}^n$ a homeomorphism onto an open subset. We consider manifolds without boundary. The component functions of the chart are its *coordinates*, written $x^1, \dots, x^n$. The superscripts are coordinate labels, not powers.

Two charts are *smoothly compatible* when both coordinate changes on their overlap are smooth. For charts (U, φ) and (V, ψ), one such change is

$$\psi \circ \varphi^{-1}\colon \varphi(U \cap V) \longrightarrow \psi(U \cap V).$$

Smooth means C^∞: all orders of partial derivatives exist and are continuous. An empty overlap imposes no condition. A *smooth atlas* is a collection of pairwise compatible charts covering M. A *smooth structure* is a maximal smooth atlas, containing every chart compatible with it. A *smooth manifold* is a topological manifold with a smooth structure.

Any smooth atlas determines a unique smooth structure. To prove this, adjoin every chart compatible with the given atlas. For two added charts ψ and θ and a point in their overlap, choose an original chart φ containing that point. Locally the transition is

$$\theta \circ \psi^{-1} = (\theta \circ \varphi^{-1}) \circ (\varphi \circ \psi^{-1}).$$

Both factors are smooth. Reversing the roles gives a smooth inverse. Thus the enlarged collection is an atlas. Any chart compatible with the enlarged collection was compatible with the original one and has already been added. This proves maximality. Every chart of

any maximal atlas containing the original atlas belongs to this enlarged collection. That maximal atlas must therefore equal the enlarged collection, proving uniqueness.

The smooth structure is extra data. On the topological line, the global charts $x(t) = t$ and $y(t) = t^3$ individually define smooth structures. Their transition $y \circ x^{-1}(s) = s^3$ is smooth, but its inverse is not differentiable at zero. Along $s = 1/k^3$, its difference quotient is $(1/k)/(1/k^3) = k^2$, which is unbounded as $k \to \infty$. Hence the two charts do not belong to one smooth atlas. The resulting manifolds will nevertheless be diffeomorphic, by a map different from the identity.

We also fix the index convention used in later calculus. A repeated upper and lower index is summed from 1 to n, unless another range is stated. For a row of coefficients a_i and a column of components X^i, write $a_i X^i = \sum_{i=1}^{n} a_i X^i$. The index position distinguishes transformation rules. It does not mean that a metric has been used. For example, a linear coordinate change and the corresponding rules are

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad x' = Jx, \qquad X' = JX, \qquad a' = aJ^{-1}, \qquad a'X' = aX.$$

The last equality follows by cancelling $J^{-1}J$. Lower coordinate labels x_i in some prerequisite notes become x^i here. This relabelling does not raise the index of a linear functional.

7.2 Spheres, projective spaces and products

Write a point of the unit sphere $S^n \subset \mathbb{R}^{n+1}$ as (y, t), where $y \in \mathbb{R}^n$ and $t \in \mathbb{R}$. Let $N = (0, 1)$ and $S = (0, -1)$ be its north and south poles. Stereographic projection gives charts

$$\varphi_N(y, t) = \frac{y}{1 - t}, \qquad \varphi_S(y, t) = \frac{y}{1 + t}$$

on $S^n \setminus \{N\}$ and $S^n \setminus \{S\}$ respectively. Both have image $\mathbb{R}^n$. The projection is shown in a two-dimensional slice.

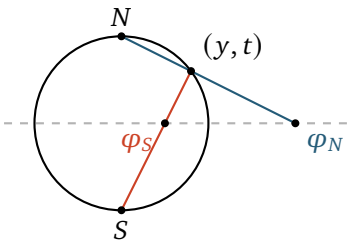


Figure 7.1: The ray from the omitted pole meets the equatorial coordinate plane. The two projections describe the same sphere away from their respective poles.

Proposition 7.1 (Standard atlases). *For $n \geq 1$, the sphere S^n and real projective space $\mathbb{R}\mathrm{P}^n$ are smooth n-manifolds. Products of smooth manifolds have smooth product atlases. In particular, $T^2 = S^1 \times S^1$ is a smooth surface.*

Proof. First, the inverse north chart is

$$\varphi_N^{-1}(u) = \left(\frac{2u}{1 + |u|^2}, \frac{|u|^2 - 1}{1 + |u|^2} \right).$$

Its squared norm is one. Substitution in $y/(1 - t)$ gives u. Conversely, for $(y, t) \neq N$, the relation $|y|^2 = 1 - t^2$ gives $|y/(1 - t)|^2 = (1 + t)/(1 - t)$. Substituting this into the displayed

formula returns (y, t). Thus it is the continuous inverse. The inverse south chart changes the sign of the last component. On the overlap, substitution gives $\varphi_S \circ \varphi_N^{-1}(u) = u/|u|^2$ for $u \neq 0$. This map is smooth and is its own inverse. The sphere is Hausdorff and second-countable as a subspace of Euclidean space. These charts therefore define its smooth structure.

Next, use Lecture 2's quotient $q\colon S^n \to \mathbb{R}\mathrm{P}^n$ by antipodes. The map q is open: the inverse image of the image of an open set W is $W \cup (-W)$. Distinct two-point orbits have disjoint small antipodally invariant neighbourhoods, proving Hausdorffness of the quotient. Images of a countable sphere basis give a countable quotient basis.

Write a class of unit vectors as $[X^0 : \cdots : X^n]$. A nonzero vector represents the class of its normalisation. Multiplying it by a nonzero scalar leaves the same antipodal class, so the notation also describes its line through zero. The open patch U_i consists of classes with $X^i \neq 0$. Its coordinates and inverse are

$$\varphi_i([X]) = (X^0/X^i, \ldots, \widehat{X^i/X^i}, \ldots, X^n/X^i), \qquad \varphi_i^{-1}(u) = [u^0 : \cdots : 1 : \cdots : u^n],$$

with the 1 in position i. The remaining coordinate labels omit i. The ratios are invariant under antipodes and continuous upstairs. Since q is open, its restriction over U_i is a quotient map, so the ratios are continuous on U_i. For the inverse, normalise the displayed representative vector and apply q. These continuous maps are inverse: normalising does not change ratios, and the ratios recover the original vector up to a nonzero scalar. On $U_i \cap U_j$, divide every component of the representative with ith component 1 by its nonzero jth component, then omit component j. All new coordinates are smooth rational functions with nonzero denominator. The same description with i and j exchanged gives the inverse transition.

The $n + 1$ patches cover, completing the projective-space assertion.

Finally, take products of charts. Their coordinate changes are products of smooth maps, so they are smooth. Finite products preserve Hausdorffness and second countability: products of the respective basis elements give a countable basis. This proves the product assertion and the torus case. ∎

7.3 Smooth maps and the chain rule

Let M and N be smooth manifolds of dimensions n and m. A continuous map $f\colon M \to N$ is *smooth* if, near every $p \in M$, its coordinate expression is smooth. For charts φ near p and ψ near $f(p)$, this expression is

$$\widehat{f} = \psi \circ f \circ \varphi^{-1}\colon \varphi(U \cap f^{-1}(V)) \longrightarrow \psi(V).$$

The domain is open because f is continuous. A *diffeomorphism* is a bijective smooth map with a smooth inverse.

> **Proposition 7.2** (Independence of coordinates)**.** *Smoothness is independent of the chosen charts. Compositions of smooth maps are smooth. Their coordinate Jacobians obey the chain rule.*

Proof. With different charts $\widetilde{\varphi}$ and $\widetilde{\psi}$, the new expression, on its common local domain, is

$$\widetilde{\psi} \circ f \circ \widetilde{\varphi}^{-1} = (\widetilde{\psi} \circ \psi^{-1}) \circ \widehat{f} \circ (\varphi \circ \widetilde{\varphi}^{-1}).$$

The outer maps are smooth transitions. This proves independence at each point. For a further smooth map $g\colon N \to P$, use the same middle chart on N. The coordinate expression of $g \circ f$ is $\widehat{g} \circ \widehat{f}$, and the Euclidean chain rule gives

$$D(\widehat{g} \circ \widehat{f})_x = D\widehat{g}_{\widehat{f}(x)}\, D\widehat{f}_x.$$

Both smoothness of the composition and the Jacobian formula follow. ∎

Here $D\widehat{f}_x$ denotes the $m \times n$ matrix of first partial derivatives. In component notation, the same formula is

$$\frac{\partial (g \circ f)^a}{\partial x^i} = \frac{\partial g^a}{\partial y^b} \frac{\partial f^b}{\partial x^i},$$

with the repeated index b summed. These are coordinate expressions; Lecture 8 defines the differential without choosing coordinates.

For the two smooth structures on the line above, the map $f(t) = \sqrt[3]{t}$ from the x-structure to the y-structure has coordinate expression $y \circ f \circ x^{-1}(s) = s$. Its inverse has the identity coordinate expression too. Thus it is a diffeomorphism, although the identity map between those structures is not a diffeomorphism.

7.4 Regular level sets and embeddings

A subset S of a smooth N-manifold M is an *embedded d-submanifold* if, with the subspace topology, it satisfies the following condition. At every $p \in S$ there is an ambient smooth chart (U, φ) containing p such that

$$\varphi(S \cap U) = \varphi(U) \cap (\mathbb{R}^d \times \{0\}).$$

The restricted coordinates give its smooth structure.

For a smooth map $f : W \to \mathbb{R}^k$, where $W \subseteq \mathbb{R}^N$ is open, a point $p \in W$ is *regular* when Df_p is onto. A *regular value* is a point $c \in \mathbb{R}^k$ such that every point of $f^{-1}(c)$ is regular. A value outside the image is regular vacuously.

Theorem 7.3 (Regular level sets · *proved given the implicit function theorem*)**.** *If c is a regular value of f, its nonempty level set $f^{-1}(c)$ is an embedded submanifold of W of dimension $N - k$.*

Proof. Fix $p \in f^{-1}(c)$. Surjectivity gives a nonsingular $k \times k$ minor of Df_p. Permute the input coordinates and write them as $(s, t) \in \mathbb{R}^{N-k} \times \mathbb{R}^k$, with this minor in the t coordinates. The smooth implicit function theorem, cited in the form of Tu, Theorem B.2, gives open neighbourhoods in which

$$f(s, t) = c \quad \Longleftrightarrow \quad t = h(s)$$

for a smooth function h. In this neighbourhood the ambient coordinate change $\Phi(s, t) = (s, t - h(s))$ has smooth inverse $\Phi^{-1}(s, u) = (s, u + h(s))$. It sends the level set to $u = 0$. These are the required ambient charts. On their overlaps, the restricted transitions are smooth, because they are restrictions of smooth ambient transitions followed by projection onto the first $N - k$ components. The subspace is Hausdorff and second-countable, inherited from W. Thus these charts define the claimed embedded manifold. ∎

This is the Euclidean case of Tu, Theorem 9.9, pp. 105–106. For the sphere, use $f(x) = |x|^2$. At $p \in S^n$, its derivative is $Df_p(v) = 2p \cdot v$, which takes the value 1 on $v = p/2$. Hence 1 is regular and the sphere has dimension $(n + 1) - 1 = n$. The resulting structure agrees with the stereographic one, since the stereographic maps and their inverses are smooth ambient formulas on their respective domains.

For the orthogonal group, the constraint takes values in the vector space Sym_n of real symmetric matrices:

$$f : \mathbb{R}^{n \times n} \to \mathrm{Sym}_n, \qquad f(A) = A^T A.$$

The target has dimension $n(n+1)/2$, since its diagonal and entries above the diagonal determine it. Expanding $f(A+tH)$ to first order in the real parameter t gives

$$Df_A(H) = A^T H + H^T A.$$

At $A \in \mathrm{O}(n)$, any symmetric matrix S is the image of $H = AS/2$, since $A^T A = I$. Thus I is a regular value and $\dim \mathrm{O}(n) = n^2 - n(n+1)/2 = n(n-1)/2$. Taking all n^2 matrix entries as independent target constraints would incorrectly count the repeated off-diagonal equations.

A smooth map is an *immersion* if its coordinate Jacobian is injective at every point. An immersion is an *embedding* if it is also a homeomorphism onto its image with the subspace topology. Chart changes multiply the Jacobian by invertible matrices, so the immersion condition is independent of charts. An injective immersion need not be an embedding. Fix an irrational number $\alpha > 0$ and consider

$$\gamma : \mathbb{R} \longrightarrow T^2, \qquad \gamma(t) = (e^{\mathrm{i}t}, e^{\mathrm{i}\alpha t}).$$

Equal images require a difference $2\pi m$ with $m \in \mathbb{Z}$ and $\alpha m \in \mathbb{Z}$. Irrationality forces $m = 0$, proving injectivity. The first circle component has nonzero derivative, proving immersion. The square model shows the repeated returns of its image.

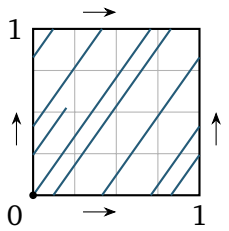


Figure 7.2: A finite segment of the line of slope $\sqrt{2}$ in the square quotient $\mathbb{R}^2/\mathbb{Z}^2$. Each exit continues at the identified opposite edge.

There are unbounded positive integers q_j with distance from αq_j to an integer tending to zero. To see this, divide $[0,1)$ into N equal intervals and place the $N+1$ fractional parts of $0, \alpha, \ldots, N\alpha$ into them. Two occupy the same interval. Their difference gives $1 \le q \le N$ with distance less than $1/N$. Such choices must be unbounded along a subsequence, since each of finitely many nonzero irrational multiples has positive distance from the integers. Consequently $\gamma(2\pi q_j) \to \gamma(0)$, although $2\pi q_j$ does not tend to zero. The inverse from the image is not continuous. Thus γ is not an embedding. Lecture 9 describes motion along this curve by prescribing its velocity at every point.

7.5 The qubit's projective state space

A pure qubit state is a nonzero column $(a, b) \in \mathbb{C}^2$, with two columns representing the same state when they differ by a nonzero complex scalar. Normalising the column leaves precisely the usual overall phase freedom. The quotient is the *complex projective line* $\mathbb{C}\mathrm{P}^1$, with points written $[a : b]$. Give it the quotient topology from $\mathbb{C}^2 \setminus \{0\}$. The patches with $a \neq 0$ and $b \neq 0$ have coordinates

$$z = b/a, \qquad w = a/b.$$

Their inverses are $z \mapsto [1 : z]$ and $w \mapsto [w : 1]$. The quotient projection is open because saturating an open set takes the union of its images under nonzero scalar multiplications.

Thus the patches are open and the ratios define continuous charts. On the overlap $z \neq 0$, the transition is $w = 1/z$. For $z = u + \mathrm{i}v$, this is the real smooth map

$$(u, v) \longmapsto \left(\frac{u}{u^2 + v^2}, \frac{-v}{u^2 + v^2} \right).$$

It is its own inverse. The missing point of either chart is the origin of the other chart, as the sphere picture indicates. Lecture 20 differentiates this transition to describe vectors attached to the sphere.

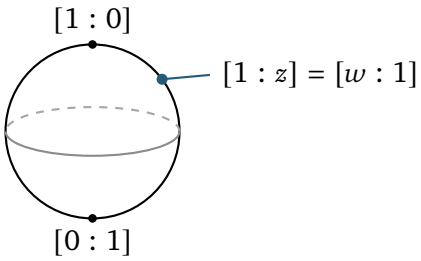


Figure 7.3: The north state $[1 : 0]$ has $z = 0$ and the south state $[0 : 1]$ has $w = 0$. Both coordinates are available away from the poles, with $w = 1/z$.

Example 7.4 (Projective states and the Bloch sphere)**.** The map $B\colon \mathbb{C}\mathrm{P}^1 \to S^2$ to the unit sphere is given by

$$B([a : b]) = \frac{1}{|a|^2 + |b|^2} \big(2\,\mathrm{Re}(\bar{a}b),\ 2\,\mathrm{Im}(\bar{a}b),\ |a|^2 - |b|^2\big).$$

Multiplying a and b by a common nonzero scalar multiplies numerator and denominator by the same positive number, so the map is well-defined. The formula on nonzero columns is continuous with nonvanishing denominator. Lecture 2's quotient criterion therefore makes B continuous. Its squared norm is one because $4|a|^2|b|^2 + (|a|^2 - |b|^2)^2 = (|a|^2 + |b|^2)^2$. In the z chart it becomes

$$B([1 : z]) = \frac{(2\,\mathrm{Re}\,z,\ 2\,\mathrm{Im}\,z,\ 1 - |z|^2)}{1 + |z|^2}.$$

For a sphere point (X, Y, Z) away from the south pole, the inverse coordinate is $z = (X + \mathrm{i}Y)/(1 + Z)$. Away from the north pole it is $w = (X - \mathrm{i}Y)/(1 - Z)$. These formulas are inverse on their patches and agree by $w = 1/z$ on the overlap. Indeed, the displayed z formula for B is exactly the inverse south stereographic chart, with real coordinates $(\mathrm{Re}\,z, \mathrm{Im}\,z)$. On the w patch it is the inverse north chart with coordinates $(\mathrm{Re}\,w, -\mathrm{Im}\,w)$. The already proved stereographic identities establish both inverse compositions. Hence B is a homeomorphism. In particular, $\mathbb{C}\mathrm{P}^1$ is Hausdorff and second-countable, inherited through this homeomorphism from the sphere. Its compatible charts therefore make it a smooth surface. The same formulas are smooth in both directions, so B is a diffeomorphism.

Take angles $0 \le \theta \le \pi$ and $0 \le \phi < 2\pi$. For the normalised representative $(a, b) = (\cos(\theta/2), e^{\mathrm{i}\phi}\sin(\theta/2))$, the z coordinate, when $\theta < \pi$, is $z = \tan(\theta/2)e^{\mathrm{i}\phi}$. At $\theta = \pi$ the state is $[0 : 1]$, with $w = 0$. The sphere image for the full angle range is $(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. Thus the two-component state and the sphere point describe the same pure state, with no overall phase coordinate.

Self-check. Use the regular level set theorem to show that $\mathrm{U}(n)$ is a real manifold of dimension n^2. Specify the real target of the constraint.

Answer. Use $f(A) = A^\dagger A$ on $\mathbb{C}^{n\times n} \cong \mathbb{R}^{2n^2}$ with target the real vector space of Hermitian matrices. That target has n real diagonal entries and one complex entry for each pair above the diagonal, so its real dimension is $n + 2\binom{n}{2} = n^2$. Differentiating with respect to a real parameter gives $Df_A(H) = A^\dagger H + H^\dagger A$. At a unitary A, every Hermitian S is reached by $H = AS/2$. Thus I is a regular value and its level set has dimension $2n^2 - n^2 = n^2$. Treating this as a complex-linear constraint would be incorrect: conjugation is part of the formula.

Summary. A smooth structure is specified by charts with smooth transitions. Smooth maps are tested in these charts, and the Euclidean chain rule makes the test independent of coordinates. The implicit function theorem turns independent smooth constraints into embedded submanifolds. The qubit state space has the two projective charts z and $w = 1/z$, and is diffeomorphic to the Bloch sphere. Lecture 8 uses coordinate changes to define tangent vectors and differentials intrinsically.

Reading. Tu, *An Introduction to Manifolds*, §§5–7, Theorem 9.9 and §11; the irrational-line example is Example 15.9. Nakahara, *Geometry, Topology and Physics*, §5.1.